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Predicting Under- and Overperforming SKUs within the Distribution–Market Share Relationship

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Abstract

This research presents a retail analytics application which uses machine learning (ML) to identify and predict under- and overperforming consumer packaged goods (CPGs) using retail scanner data. Essential to measuring market performance at the SKU level is the relationship between distribution and market share (the velocity curve). We validate that ML can reproduce the velocity curve, and ML is further used to predict underperforming, in-line performing, and overperforming SKUs relative to the velocity curve, based on a range of variables (SKU features) at a point in time. Our ML approach can correctly predict 83% of SKUs as under-, in-line-, or overperforming based on their characteristics. The research analyzes 9,321 SKUs of 2,565 brands across seven product categories of CPGs which were sold in 8,117 stores from 49 different retail chains of five different retail channels located in the US states of California, New York, Texas, and Wisconsin. The retail stores comprise convenience stores, drug stores, food stores, liquor stores, and mass merchandise retail stores. The data is Nielsen retail store scanner data for the calendar year 2014. The relationship between distribution and market share is a market-wide proxy for the ratio of relative sales in a category to, for example, aggregate shelf space, a key retail productivity metric. We further find indications that the distribution of SKUs across different store sizes, the stores' category specialization, the line length of the brands, the overall performance of the parent brand, and sales consistency are the most important characteristics for the prediction of market share performance beyond the velocity curve. The methods and results presented will help CPG marketers (suppliers and retailers) understand which SKUs are under-, in-line-, or overperforming and the potential factors contributing to that performance. Optimizing assortments and portfolios is essential to decrease failure rates of individual SKUs. ML approaches can evolve to complementary support tools for such management problems.
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Introduction

Developments in the retail landscape reveal new challenges for retailers and consumer packaged goods (CPG) marketers. The growth, variety, and size of available data, complex omni-channel relationships, fast information flow, and changing shopper behaviors require modern approaches in retail analytics to create insights for decision makers (Grewal, Roggeveen, and Nordfält 2017; Kumar, Anand, and Song 2017; Marketing Science Institute [MSI] 2018). The advances in data- and computer science present opportunities for marketers to address these challenges by utilizing new approaches such as machine learning (ML) to analyze more elaborate data in a manner that

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will complement conventional practices (Cui and Curry 2005; Wedel and Kannan 2016). The present research demonstrates the predictive power of ML and how it can provide additional insights from the convex distribution – market share relationship, which is a well-known market pattern that Ailawadi and Farris (2020) recently termed the "velocity curve". Velocity refers to the ratio of a stock keeping unit's (SKU's) or brand's sales or market share to a distribution metric, such as all-commodity volume (ACV).

Velocity and associated metrics are commonly used in the CPG and retail industry, reflecting their importance to support management decisions. CPG marketers like Mondelez, Procter & Gamble and Coca-Cola have a need to efficiently manage their portfolios by prioritizing best performing and cutting inefficient SKUs (Cosgrove 2020). Leading venture capitalists in CPG brands reinforce the importance of early velocity prior to growth through distribution breadth (Begoun 2017). Experienced analysts rank the velocity metric as the third most important CPG metric after sales (e.g., market share) and distribution (e.g., ACV) (Martin 2014). Velocity curves combine these metrics into relationships and have the advantage of ordering brands and SKUs based on their velocity. These metrics also allow us to model expected velocity performance specific to a set of products or a category in a market.

In the first part of our paper, we extend the convex distribution–market share relationship and associated velocity curve to statistically explain the performance of SKUs. We employ a simple parametric formula (Reibstein and Farris 1995) to model the relationship. We establish the validity of the convex distribution–market share relationship at the SKU level and then validate ML as a new method of modeling the velocity curve at the SKU level.

We show that the velocity curve explains an average of 72% of distribution and market share variation across product categories, leaving 28% unexplained variation. This opens up the value of more elaborate SKU level data to gain further insight from the velocity curve. Extending the velocity curve, we identify and group SKUs into three classes: those that perform below, in-line with, or above expected market share (velocity curve deviations). Weighted random forest (WRF), which is a type of supervised ML, is then used to predict[2] these three classes using data related to product, brand, line, category, and distribution characteristics.

Objective:	*Use ML to predict underperforming, in-line performing, and overperforming SKUs in relation to the velocity curve, and identify potential SKU characteristics that may be associated with deviations from the velocity curve.*

We validate the ML predictions about the three SKU performance classes against a benchmark model—ordinal logistic regression (OLR)—under the same test conditions. Overall, the WRF correctly predicts on average 83 out of 100 SKUs' performance relative to the velocity curve. The approach presented here provides a powerful method for analyzing the relative velocity of the many thousands of SKUs stocked by retailers. The approach also lays a foundation for further research into the SKU attributes that signal more or less permanent versus temporary deviations from velocity curves. These insights are needed to fully evaluate the use of velocity curves in the relationship between market share and distribution performance.

General Aim of the Research and Relevance for Category and Portfolio Management

One of the most important research findings in the retailing literature is that of the relationship between distribution and market share (Farris, Olver, and De Kluyver 1989; Hirche et al. 2021; Reibstein and Farris 1995; Wilbur and Farris 2014). Distribution has a consistently convex relationship with market share, which has been demonstrated across multiple products, categories, and contexts (Farris, Olver, and De Kluyver 1989; Hirche et al. 2021; Kruger and Harper 2006; Wilbur and Farris 2014). Current research also highlights the importance of distribution for market performance (Ataman, Mela, and van Heerde 2008; Elberse and Eliashberg 2003; Kumar, Sunder, and Sharma 2015) and suggests that up to 84% of the variation in market share can be statistically attributed to distribution (Kruger and Harper 2006). While the statistical power of the relationship is strong, it is far from perfect. Literature has proposed multiple factors to account for the remaining market share variation (Bronnenberg and Dubé 2017; Farris et al. 2015; Hanssens, Parsons, and Schultz 2002; Kumar 1994), but few consistent patterns have emerged.

Market share is both a result (a product is available for purchase) and a cause of distribution (higher turnover rates make a product more likely to be adopted by more retailers). Because different retailers have different assortment sizes and strategies, stocked products enjoy enough turnover (market share) to satisfy stocking retailers, but not enough turnover to attract more retail distribution. However, the constant introduction of new products and the growth and decline of retailer firms and formats mean that the relationship between distribution and market share is rarely completely stable for long periods, especially for individual SKUs. Therefore, a major challenge for CPG marketers and retailers is to decide which SKUs to discontinue/delist and which to release/adopt to manage portfolios and category assortments. This raises the question of how we can best explain the remaining variation in market share to support decisions by product and brand managers in a highly ambiguous environment.

Wedel and Kannan (2016) advise that marketing analytics should combine ML techniques with classical model approaches to ensure a balance between black-box predictions and generalizable inferences. Cui and Curry (2005) address this approach by proposing a framework for comparing ML approaches with conventional modeling techniques, and thereby highlight the complementary (rather than competitive) nature of ML with parametric modeling. This important notion is reflected in the present research. ML may assist assortment decisions in retail practice reflecting its multidimensional nature. We validate ML as an empirical approach and complement the velocity curve through an application of ML that employs retail scanner data

[2] In this study the ML prediction is concerned with estimating the outcome of unseen (new) data, not with predictions about the future (forecasting).

to identify and predict underperforming, in-line performing, and overperforming SKUs of CPGs in new data.

From a management perspective, accurate predictions of possible market outcomes are extremely valuable. Considering the costs of marketing activities, sensible and efficient allocation of budgets is crucial for market success and profitability. There is a clear need for retail analytics to inform retail strategy because misinformed decisions can have severe consequences such as sales drops that can result in delisting or discontinuing decisions (Fisher and Vaidyanathan 2012). Typically, adding an SKU to an assortment is accompanied by the delisting of nonperforming SKUs. The important question is which SKU should be added and which should be dropped. Advanced analytical methods, including ML, can support such management decisions. This is why the seemingly simple but important generalization of the relationship between distribution and market share presents a valid foundation for supplementary analytics, as undertaken in the present research. The ML application that is used by the present research to predict underperforming, in-line performing, and overperforming SKUs should be understood as one of many possible techniques that can be used to identify SKUs that are likely to experience some unusual market performance, and therefore require subsequent examination of potential causes.

A CPG supplier can use this ML application for their product lines/portfolios when selling their CPGs through retail stores. The velocity curve will indicate what market share to expect in relation to distribution (or vice versa), and the ML application predicts whether the velocity curve applies (in relation to expected normal market share) or whether individual SKUs are likely to be underperforming or overperforming. Such predictions can serve as a benchmark for CPG suppliers. In contrast, retailers can support their listing and delisting decisions by using such ML applications to make predictions for single SKUs in an assortment. If a retailer plans to delist an SKU in only a subset of all stores, this will affect (i.e., decrease) the distribution breadth of this SKU in the market. With the velocity curve, we can expect a certain market share given the distribution of this SKU, in addition, ML predicts underperformance, in-line performance, and overperformance based on SKUs deviating from the velocity curve. The ML application is concerned with the accurate prediction of performance using a wide range of SKU specific product, brand, category, and distribution variables. The precise reasons for this performance (causality) can be examined through subsequent (post-hoc) analyses. ML does not assume that such causal explanations are available from the data. Post-hoc analyses such as controlled experiments, descriptive analysis, pairwise methods or partial dependence plots (Zhao and Hastie 2019) will inform further assessments and decisions on a case-by-case basis.

Foundation for Predicting SKU Performance

Several studies across different markets and product categories have explored the pattern of the velocity curve (Borin, Van Vranken, and Farris 1991; Farris, Olver, and De Kluyver 1989; Kruger and Harper 2006; Verbeke, Clement, and Farris 1994; Wilbur and Farris 2014). Wilbur and Farris (2014) extended previous research to investigate the relationship at the SKU level, as opposed to the brand or brand-line level, and found most patterns to be convex. It is generally accepted that in CPG product categories, the pattern appears as convex and increasing; many low-share and medium-share brands bundle at the lower ends of the relationship, and fewer high-share brands distribute exponentially at the higher ends.

Previous studies have explored possible approaches to modeling the relationship between distribution and market share. For example, Verbeke, Clement, and Farris (1994) developed a set of formulae that interdependently account for preference, in-store promotion, compromised and uncompromised demand, and stocking rules related to store class to reflect push and pull dynamics in the market. More recently, Wilbur and Farris (2014) tested a parametric quadratic function across various category and brand conditions. However, besides some of the limitations such as the data not including some retail channels or private label (PL) brands, the functional form of their model allows only for linear (if the parameter of the quadratic term is exactly zero), fully convex, fully concave, or parabola curves. In addition, the high maximum distribution values also indicate a limited sample of only two retailers.

Reibstein and Farris (1995) modeled this velocity curve using a simple parametric formula. Their model has the advantages of being simply formulated and easily applicable, as well as being flexible in adopting mixed curve patterns (e.g., s-shaped), therefore making it a likely candidate for the present research to establish the baseline for further predictive analysis. In contrast to other studies (e.g., Wilbur and Farris 2014), our study includes PL brands to reflect a competitive market that includes retail brands. PLs can have a strong influence on the curve pattern because PL SKUs are more likely to have limited distribution with relatively high market share. This is due to them being sold exclusively in only several stores. An argument for excluding PLs is that the estimated velocity curves are more likely to be achievable by manufacturer brands. Our perspective includes the potential use of velocity curve by retailers, so we have opted to include PLs in our analysis.

The relationship between distribution and market share raises further questions about SKUs that substantively deviate from the velocity curve. Some of this deviation may be due to SKUs exceeding or falling below retail turnover thresholds with the expectation that the SKU market share and distribution values will move to fit the curve over time. Other deviations may be more or less permanent (e.g., PLs). There are indications that certain product and distribution characteristics explain differing market performance (e.g., Venkatesan et al. 2015). Ailawadi and Farris (2020) note that velocity curves can be guides to evaluating the sustainability of SKU portfolios and distribution levels of individual SKUs. However, these researchers also note that there may be good reasons explaining why SKUs more or less permanently occupy positions off the curve (Ailawadi and Farris 2020). To the best of our knowledge, no study has examined a large set of general characteristics that are supposedly associated with deviations from the velocity curve and attempted to predict these using ML. We aim to fill this gap so that we can further study which factors might produce more or less perma-

nent deviations, and which might be indicators of an entire class of SKUs that are overperforming or underperforming, and are hence candidates for adoption or delisting decisions.

Limitations of Theoretical Development

The main reason ML is potentially useful for predicting SKU performance is the limited theory on the drivers of overperformance and underperformance of SKUs in terms of market share relative to distribution at the SKU level. While such theory is limited, theory can indeed play an important role in determining the potential relevance of factors (variables) for inclusion in ML. While this limited theory may be considered a weaker form of theory, it nonetheless plays a role in the ML process.

Given that the velocity curve and SKU performance relative to that curve is fairly unexplored, beginning with Wilbur and Farris (2014) only six years ago, there is not a great deal of literature to support theory development. However, research can view velocity curves for a particular category and universe of retailers as the collective category assortment decisions of the retailers. Category management and assessment of SKU performance are established practices that can provide some insight into potential causes of some SKUs being more or less likely to be included in a retailer's assortment for a given sales volume.

One way to describe overperforming and underperforming SKUs is that they have characteristics that prevent fast sellers from achieving expected distribution, thereby inflating velocity and in the case of lower-velocity SKUs, being able to achieve and sustain higher distribution than otherwise expected. These characteristics provide relevant variables to be included into the ML prediction. While there are many potential characteristics, we provide suppositions about some potential reasons of overperformance and underperformance.

Overperforming SKUs (i.e., those above [left of] the velocity curve) generate lower than expected distribution levels for a given market share value (or vice versa), potentially because of the following reasons:

1. SKUs recently introduced into the market with strong market demand may not have had enough time to be reviewed and accepted by many retail buyers (time in market). There needs to be time for demand estimation (Fisher and Vaidyanathan 2014) where the retailer can assess the SKU and its fit with its portfolio, and for manufacturers to communicate the benefits of the SKU (Desai 2000).
2. PL SKUs are restricted to a subset of outlets and are likely to receive particularly high merchandising emphasis from those retailers (brand size and type). Farris (2004) suggests that PL brands are limited in distribution because of retailer exclusivity, and therefore can benefit from lower retail selling prices and more retailer support.
3. Some large retail stores may allocate relatively more shelf space to big pack sizes than small pack sizes, while small retail stores may have a limited assortment with predominantly small pack sizes. Big pack size SKUs may benefit from this circumstance at relatively low distribution levels (store size and pack size). Therefore, retailers stocking preferences need special consideration (Shah, Kumar, and Zhao 2015). There are indications that SKUs listed with small retailers in small stores (limited shelf space) are at particular risk of being replaced (Hristakeva 2020). Pack size is an important criteria for retailers regarding shelf space allocation (Borin, Farris, and Freeland 1994), and store size moderates the effect of product characteristics on shelf space elasticity (Eisend 2014, p. 168).

Underperforming SKUs (i.e., those below [right of] the velocity curve) generate higher than expected distribution levels for a given market share value (or vice versa), potentially because of the following reasons:

1. SKUs supplied by direct-to-store delivery or independent wholesalers who tailor inventories to match slower sales rates, thus maintaining sustainable gross margin return on inventory investment (supply–demand efficiency and sales consistency), may be resistant to delisting despite lower market share, particularly for some product categories (Cadeaux 1994). This suggests the relationship between the wholesaler and retailer and how they work together is of considerable importance (Waller et al. 2010).
2. SKUs may have smaller shelf footprints (store penetration, retail channel penetration). SKUs that have fewer store listings, particularly in small stores, are more likely to have below expected market share, in line with past literature (e.g., Farris 2004; Shah, Kumar, and Zhao 2015). At an in-store level, SKUs that take up less shelf space, through smaller pack sizes, can result in lower sales (Waller et al. 2010).
3. SKUs in the decline stage of their life cycle (time in market), suffering from declining sales volume but not necessarily lower margin, may be targeted by retail buyers for delisting (Alexander 1964; Davies 1994), but they may not yet have found acceptable substitutes for these SKUs allowing replacement and are therefore selling off inventories (Avlonitis 1987).

In addition, all of the above potential reasons for the underperformance and overperformance of SKUs may vary by product category, channel, or market condition.

Data and Algorithms

This study employs syndicated retail store scanner data provided by Nielsen via the Kilts Center for Marketing at the University of Chicago Booth School of Business.[3] Data cleaning, deduplication, transformation, and aggregation were required to prepare analysis. R open-source software and pro-

[3] Derived based on data from the Nielsen Company (US), LLC and marketing databases provided by the Kilts Center for Marketing Data Center at the University of Chicago Booth School of Business. The conclusions drawn from the Nielsen data are those of the researchers and do not reflect the views of Nielsen. Nielsen is not responsible for, had no role in and was not involved in analyzing and preparing the results reported herein.

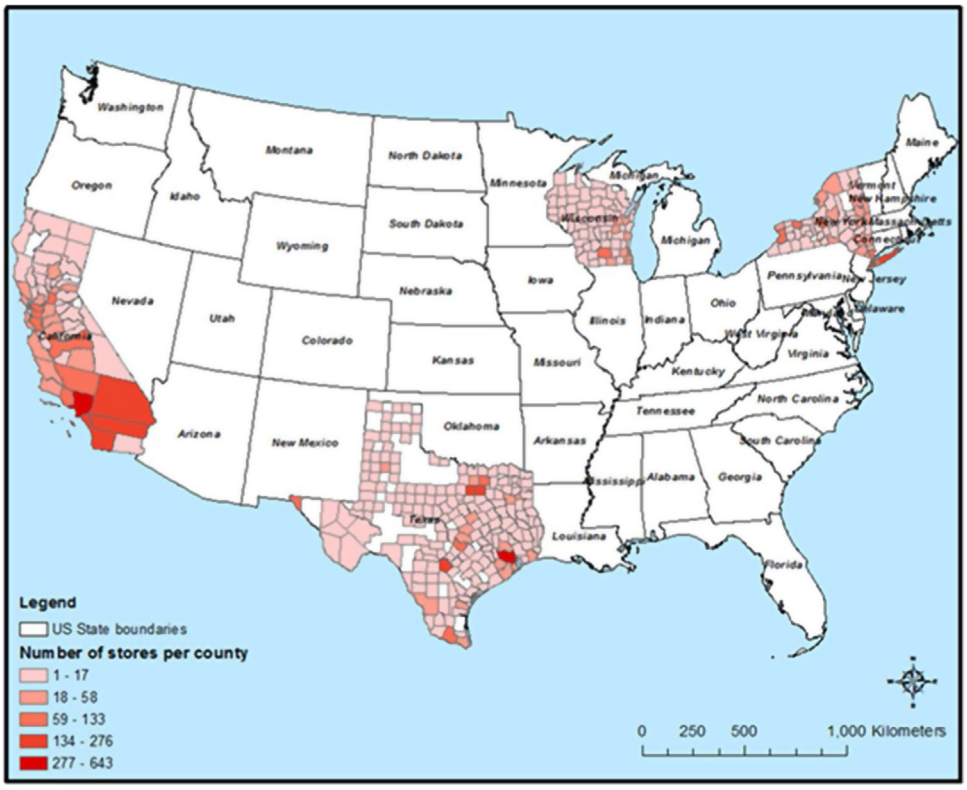


Fig. 1. Coverage of sampled United States stores by states and counties.

Source: own illustration.

gramming language was used for data processing and analysis (R-Core-Team 2019).

Retail Stores

The retail store sample includes all stores in California, Texas, New York, and Wisconsin that agreed to provide data to Nielsen and sold any of the SKUs under investigation. The selected states have high overall store coverage and high geographical dispersion, which is also evident when visualizing the distribution of the states' counties in which the sample stores are located (see Fig. 1).

Most stores in the sample are located in California, followed by Texas, New York, and then Wisconsin. Most of the stores in the sample are drug stores (40%). Approximately one-third of the total sample is composed of mass merchandise retail stores (30%). Food stores (28%) have the highest variety of different retail chains. Convenience and liquor stores have very low representation, at two percent and one percent, respectively. Overall, the final sample has 8,117 stores from 49 different retail chains.

Product Categories

The study involves 9,321 SKUs of 2,565 brands across seven product categories of CPGs that were worth US$1.6 billion in retail sales in 2014. The sampling includes a range of a-priori selected product categories. Table 1 summarizes relevant category characteristics. A comprehensive list of this sample's CPG departments and sales ratios is available in Web Appendix D.

Metrics of Distribution and Market Performance

There is no standard for the best definition and calculation of most marketing metrics. However, there are some well-established metrics that are widely accepted (Farris et al. 2015). In general, metrics must serve a purpose, and the definition and calculation of these metrics should depend on the contextual focus of the analyses, market, and time.

Market Share

One of the most important and commonly used brand-size-related performance measures in marketing is market share. In this study, we obtain the following formula.

$$\%MS_{i_k w} = 100 \times \frac{\sum_{s=1}^{p} 1_{isw} Sales_{isw}}{\sum_{s=1}^{p} Sales_{ksw}} \quad (1)$$

[4]

[4] The weekly market share *MS* (1) as a percentage of SKU i in category k in week w equals the share between the sum of all SKU i's *Sales* ($-value) of all $s = 1, \ldots, p$ indexes stores where SKU i was sold (dividend) and the sum of all category k *Sales* in p indexes stores (divisor), multiplied by 100. The coefficient 1_{isw} indicates that SKU i was sold in store s, and becomes 0 if it was not.

Table 1
Product category sample statistics.

Category	SKUs		Brands	SKUs per brand	PL SKUs	Max. market share of SKUs	Max. distribution of SKUs	Median unit price
	Count	Share (%)	Count	Average count	Share (%)	% of $	%Wtd. Dist.	$/unit
Imported dry table wine	3,527	37.8	1,628	2.2	1.8	3.1	75.5	13.38/l
Macaroni pasta	1,941	20.8	194	10.0	43.2	4.7	83.8	0.10/oz
Packaged and bulk rice	1,030	11.1	218	4.7	39.8	4.7	70.6	0.08/oz
Toilet tissue	995	10.7	87	11.4	54.1	4.2	87.9	0.59/ct
Olive oil	896	9.6	214	4.2	37.7	3.6	62.3	0.34/oz
Shelf-stable tuna	487	5.2	92	5.3	40.2	10.2	74.1	0.34/oz
Butter	445	4.8	132	3.4	48.8	8.5	78.4	0.30/oz
Total sample	9,321	100.0	2,565	3.6	27.9	5.6	76.1	n/a

Based on Eq. (1) we further calculate Eq. (2) for the final market share variable.

$$\%MS_{i_k y} = \frac{1}{52}\left(\sum_{w=1}^{52} \%MS_{i_k w}\right) \tag{2}$$

[5]

The distribution-related metrics (3) and (4) are based on a similar calculation approach.

Distribution

This study uses weighted distribution (Wtd. Dist.) as the metric for distribution of individual SKUs. Wtd. Dist. is a hybrid of ACV and product category volume (PCV). Like ACV we weight the Wtd. Dist. metric by the annual total revenue of stores, but within the universe of stores that sell the SKU's category. We redefine the universe for ACV to equal the sub-universe of stores that account for 100% PCV.

$$\%Wtd.Dist._{iw} = 100 \times \frac{\sum_{s=1}^{p} 1_{isw} Revenue_{sw}}{\sum_{s=1}^{p} 1_{ksw} Revenue_{sw}} \tag{3}$$

[6]

The final weighted distribution variable evolves from Formula (4).

$$\%Wtd.Dist._{i_k y} = \frac{1}{52}\left(\sum_{w=1}^{52} \%Wtd.Dist._{iw}\right) \tag{4}$$

[7]

[5] The average weekly market share percentage $\%MS$ of SKU i in category k equals its average weekly market share $\%MS$ of all w weeks for year y.

[6] The percentage of Wtd. Dist. for SKU i in week w is the sum of all $s = 1, \ldots, p$ indexes store's *Revenue* that sold SKU i, divided by the sum of all $s = 1, \ldots, p$ indexes store's *Revenue* that sold SKU i's product category k, multiplied by 100. Again, 1_{isw} and 1_{ksw} are the coefficients that indicate whether store s has sold SKU i, and whether store s has sold any SKU of category k, respectively.

[7] The average weekly weighted distribution %Wtd. Dist. of SKU i in category k for year y equals its average weekly weighted distribution %Wtd. Dist. of all w weeks in year y.

Modeling the Velocity Curve Relationship

Having established the %MS and %Wtd. Dist. metrics, we implement the model and the respective assumption tests. A nonlinear least squares (NLS) regression model based on Reibstein and Farris's (1995) market share Eq. (5) was executed using the *nlstools* package in R (Baty and Delignette-Muller 2015; Baty et al. 2015). Compared with ordinary least squares regression, NLS requires iterative numerical approximations to optimize the parameters (Nash and Varadhan 2011).

$$MS_{ik} = \frac{\beta_{0_k} * Wtd.\,Dist._{ik}^{\beta_{1_k}}}{(100 - Wtd.\,Dist._{ik})^{\beta_{2_k}}} \tag{5}$$

[8] where $Wtd.\,Dist., MS \in [0, 100]$; $\lim_{n\to\infty} Wtd.\,Dist._n = 100$; $\beta_0, \beta_1, \beta_2 \in \mathbb{R}_+$.

The NLS regression is the foundation for the dependent variable of SKUs' market share deviation from their expected market share. The deviation represents the above or below expected market share performance of individual SKUs. The formula constitutes as follows.

$$MSD_{ik} = \hat{u}_{ik} = MS_{ik} - \frac{\hat{\beta}_{0_k} Wtd.Dist._{ik}^{\hat{\beta}_{1_k}}}{(100 - Wtd.Dist._{ik})^{\hat{\beta}_{2_k}}} \tag{6}$$

[9]

[8] The market share (MS) for every i^{th} SKU in every k^{th} product category equals the parameter β_0 multiplied by the SKU's weighted distribution (*Wtd. Dist.*) raised to the power of parameter β_1, divided by the subtraction of 100 minus weighted distribution (*Wtd. Dist.*) raised to the power of parameter β_2. Market share and weighted distribution are restricted real numbers between 0 and 100. *Wtd. Dist.* has an upper limit of 100 and is not defined at 100. All parameters are nonnegative real numbers. Even though estimated with MS as the dependent variable the model is theoretically bi-causal with *Wtd. Dist.*.

[9] The market share deviation of every i^{th} SKU in every k^{th} product category equals the estimated error ($\hat{u}$), which equals the subtraction of the NLS regression's estimated market share from the actual market share (MS). $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ represent the fitted parameters for each product category.

Underperforming, In-line Performing, and Overperforming SKUs

To simplify the managerial decision making concerning the market share deviation of SKUs from the velocity curve, we suggest finding some threshold that distinguishes underperforming and overperforming SKUs, as well as SKUs that are in-line with the velocity curve. SKUs that are close to the curve have been sufficiently explained by the relationship between distribution and market share. Management wants to avoid or delist SKUs below the curve (underperforming) given their distribution, aiming at SKUs to be part of the top end of the market above the curve (overperforming). We propose at least two approaches to define underperforming, in-line performing, and overperforming SKUs:

1. statistically driven (prediction interval),
2. managerially relevant (top/bottom ten percent).

The first approach (1) uses the prediction interval in which future observations should fall with some probability. SKUs above, within, or below the interval can be labeled as underperforming, in-line performing, and overperforming. The second approach (2) distinguishes the ten percent of SKUs in each product category with the highest (overperformers) market share deviation and the lowest (underperformers) market share deviation. The remaining 80% are in-line performers. Both approaches will be tested and their feasibility for ML analysis assessed.

Prediction with WRF

To address the research objective, we use WRF, which is a sophisticated ML algorithm,[10] to predict the three classes of SKU performance. The analysis involves 57 input variables that the ML algorithm uses to train itself to predict the output, which is a categorical variable that indicates the three levels of SKU performance (underperforming, in-line performing, overperforming). Further details about the variables' type, definition, and measurement can be found in Appendix A. The variables are broadly related to product and distribution characteristics and potentially related to overperformance and underperformance of SKUs in retail stores. As discussed, theory suggests using the following constructs to predict underperforming, in-line performing, and overperforming SKUs: product, brand, time in market, supply–demand efficiency (including sales consistency), and channel and store penetration. The variables available to capture these constructs are category, pack size, multi-pack/single-pack, price (indicators of product), brand type, number of variants and SKUs of the brand, average performance of other SKUs of the brand (indicators of brand), weeks sold overall and in different store sizes (indicators of time in market, supply–demand efficiency, and sales consistency), number and share of retail chains and channels, share of different store sizes, category specialization of stores (indicators of channel and store penetration), and various interaction variables.

WRF

We seek a suitable ML classification that can deal with predicting more than two classes (three classes in this study). Fernández-Delgado et al. (2014) evaluated 179 ML classifiers across 121 datasets and based on prediction accuracy and speed of processing, identified random forest as the best family of classifiers, followed by support vector machines, neural networks, and boosting ensembles. Further, a recent meta-analysis of 48 published studies that applied multiple ML algorithms compared prediction accuracy outcomes across logistic regression, support vector machine, decision tree, random forest, naïve Bayes, k-nearest neighbor, and artificial neural network (Uddin et al. 2019). The analysis confirmed that comparatively, random forest algorithms offer superior accuracy in a wide spectrum of settings. Instead of obtaining one prediction (simple regression tree), random forests produce many predictions and average these predictions. Random forests have good implementation and have the following minor advantages over support vector machines for our specific setting:

1. Random forests can be significantly faster to train (processing time).
2. Random forests can deal with multiple classes (as in our case), whereas support vector machines are designed for binary classification and would have to be adapted when there are more than two classes.

Traditional random forests are constructed to minimize the overall error rate and tend to focus more on the prediction accuracy of the majority class, which often results in poor accuracy for the minority class. Thus, in this study, we apply WRF to accommodate the unbalanced size of the SKU classes. This approach imposes a heavier penalty for misclassifying the minority class by assigning a weight to each class, with the minority class given a larger weight (higher misclassification cost). To address this imbalance of classes we run WRF with the weighting formula (Chen, Liaw, and Breiman 2004).

$$\omega_j = \frac{1}{k * n_j} \tag{7}$$

Classes are weighted inversely proportional to how frequently they appear in the data. The class weights are incorporated into the random forest algorithm in two places:

1. weighting the Gini criterion for finding splits,
2. "weighted majority vote" in class prediction of each terminal node.[11]

[10] The full WRF algorithm, further model diagnostics and descriptive statistics are publicly available at https://github.com/public-research/SKU-ML.git.

[11] Weighted vote of a class is the weight for that class times the number of cases for that class at the terminal node

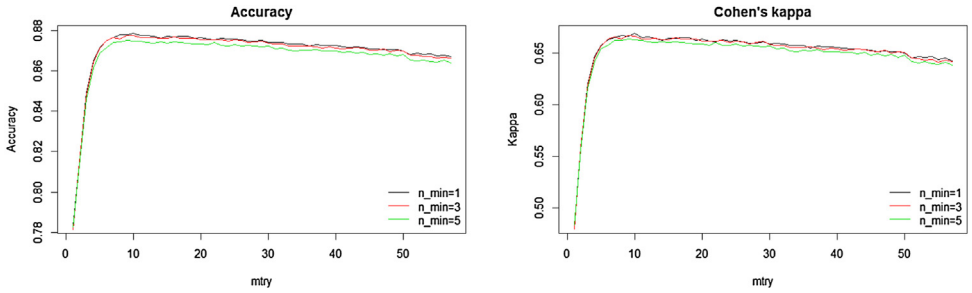


Fig. 2. Parameter tuning based on accuracy and kappa criteria.

Measuring and Evaluating the WRF

We use the most common metrics to evaluate the WRF and directly compare the results with a benchmark model, thereby following a similar approach to that used by Cui and Curry (2005). OLR serves as the benchmark because it is the most general model capable of analyzing an ordinal dependent variable. Predictions are based on the same training data, test data, and independent variables as used in the WRF. The evaluation metrics are as follows:

1. accuracy with confusion matrix,
2. Cohen's kappa,
3. area under the receiver operating characteristic (ROC) curve (AUC).

Accuracy (1) is the percentage of correctly classified instances out of all instances. A confusion matrix will provide further details about correctly and incorrectly predicted instances including the related sensitivity and specificity metrics. Cohen's kappa (2) is similar to classification accuracy, except that it is normalized at the baseline of random chance. It is a more useful measure to use in datasets that have an imbalance in the classes. According to Landis and Koch (1977), the following are the Cohen's kappa value ranges for a evaluative interpretation of the ML performance: 0.00 = no agreement; 0.01–0.20 = slight agreement; 0.21–0.40 = fair agreement; 0.41–0.60 = moderate agreement; 0.61–0.80 = substantial agreement; and 0.81–1.00 = almost-perfect agreement. AUC (3) reflects a model's ability to discriminate between positive and negative classes. An area of 1.00 represents a model that predicts perfectly. An area of 0.50 represents a model that predicts as good as random.

We use the Caret package in R to implement WRF (Kuhn 2008) along with the Ranger package for fast implementation of random forests with high dimensional data (Wright and Ziegler 2017). We use the same training (80%) and test (20%) sets as in the previous analysis. The machine's parameters are tuned based on the number of randomly selected variables (tuning parameter m_{try}), a set minimum size of the terminal node for classification problems, and the Gini impurity.[12] The overall tuning goals are maximum accuracy and kappa values.

The three main parameters in the WRF algorithm are as follows:

1. mtry (m_{try}): the number of variables randomly sampled as candidates at each split. The default value for classification problems is $m_{try} = \sqrt{p}$, where p is the number of predictors.
2. splitrule: this defines the splitting rule. For classification and probability estimation the default is "Gini".
3. min.node.size (n_{min}): the minimum number of samples within the terminal nodes. This parameter controls the complexity of the trees. Smaller node sizes allow for deeper, more complex trees, and bigger node sizes result in shallower trees. The default value for classification problems is $n_{min} = 1$.

The optimal combination of these three parameter values with maximum accuracy and kappa values is selected for the final model, as demonstrated in Fig. 2.

Variable (Feature) Importance

Breiman (2001) proposes a method for measuring variable (feature) importance referred to as "mean decrease accuracy" (MDA).[13] We use MDA in combination with repeated tenfold cross-validation to calculate the variable (feature) importance score for each variable. Here, given that the resampling method is changed from bootstrap to cross-validation, the OOB samples are replaced by cross-validation sets. The cross-validated permutation variable importance is the average of all k-fold permutated variable importance values. This process repeats five times for more stable results. Ultimately a list of the most important variables (features) for predicting SKU classes will offer insights into the management areas that most affect the predictability of SKU performance.

[12] A split rule for tree branches that aims for maximum homogeneity within subnodes and maximum heterogeneity between nodes.

[13] MDA measures which variables contribute most to a model by using out of bag (OOB) samples to estimate the OOB accuracy and then computing the difference in accuracy before and after the permutation of individual inputs. The decrease in accuracy as a result of this permutation is averaged over all trees built in the model and will be used as a measure of variable (feature) importance. OOB samples are generated from bootstrap aggregation, which represents approximately one-third of the observations not used in bootstrap samples to fit each tree.

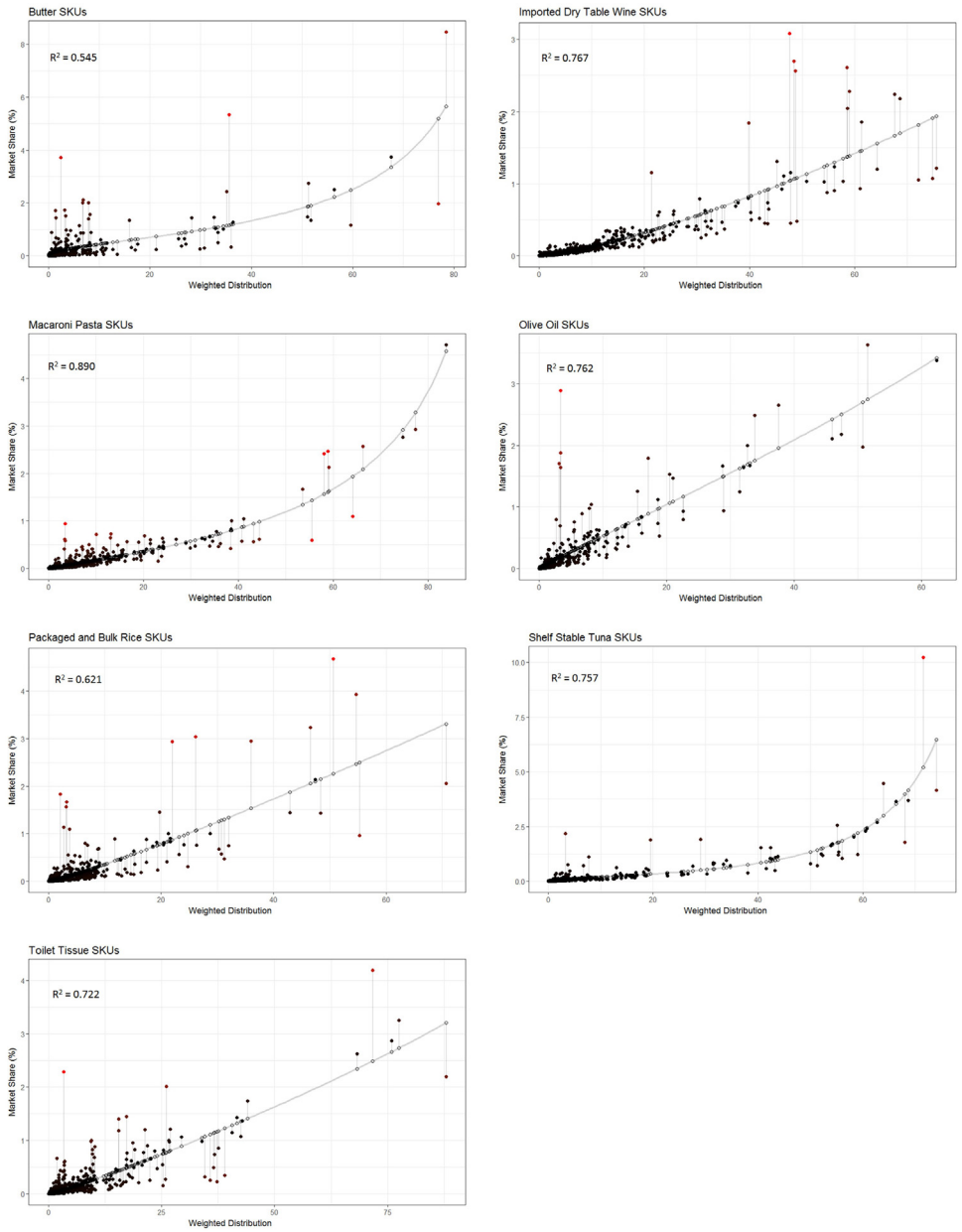


Fig. 3. Velocity curve results for product categories.
Note: The filled data points are the plotted SKUs according to the actual weighted distribution (x-axis) and market share (y-axis) for each product category. The outlined transparent points on the line represent the equivalent estimated market share for each actual data point. The straight lines connecting the actual data points with the fitted curvilinear line illustrate the deviation from the fitted line.

Results

All seven product categories have a monotonically increasing and either fully convex or partially but primarily convex form (s-curve), as expected under the velocity curve.[14] In Fig. 3, all selected categories are graphed to show the observations (SKUs) and the predicted bi-causal relationship between distribution and market share based on the velocity curve.

The NLS regressions across categories provide very good fit. The R-squared values vary from 55% for butter to 89% for macaroni pasta. On average, 72% of market share variation can

[14] The monotonicity of all modeled velocity curves is increasing. Four of the seven model curves are s-shaped (i.e., those for butter, macaroni pasta, olive oil, and shelf-stable tuna), beginning with concave intervals up to an average of 10.69% Wtd. Dist. (inflection point), then turning convex. The monotonicity criteria for a monotonic increasing interval of the function is $Wtd.Dist._1 \geq Wtd.Dist._2 \leftrightarrow f(Wtd.Dist._1) \geq f(Wtd.Dist._2)$ for the interval [0, 100]. Given that the velocity curve is a twice-differentiable function, the criteria for convex function intervals is $f''(Wtd.Dist.) \geq 0$ and for concave function intervals $f''(Wtd.Dist.) \leq 0$. If the second differentiation of the function takes both positive and negative values, it is, by definition, partially convex and concave. If the second differentiation results in one single positive value, the function is declared fully convex.

Table 2
NLS regression results for the velocity curve.

Category	Obs.	R-squared	β_0	β_1	β_2
Macaroni pasta	1,941	0.890	0.706	0.911	0.777
Imported dry table wine	3,527	0.767	0.006	1.347	0.000
Olive oil	896	0.762	0.127	0.942	0.164
Shelf-stable tuna	487	0.757	1,113.513	0.424	2.143
Toilet tissue	995	0.722	0.023	1.117	0.033
Packaged and bulk rice	1,030	0.621	0.026	1.139	0.000
Butter	445	0.545	21.769	0.471	1.108
Average	1,332	0.724	162.310	0.907	0.604
Total	9,321				

be related to weighted distribution (see Table 2). Web Appendix B presents further model diagnostics.

The model for all sampled categories shows increasing market share with increasing distribution, and the curve pattern of the bi-causal relationship is fully or partially convex. Some categories (i.e., butter, macaroni pasta, olive oil, shelf-stable tuna) have relatively small concave intervals at the lower end of distribution before turning convex (s-shaped). Fitting the bi-causal relationship between distribution and market share can be considered successful and confirms the general convex and increasing curve pattern. The successfully modeled relationship forms the basis for further investigation into the deviations from that line.

The research objective focuses on the prediction of SKU classes with ML. Aggregating many decision trees using methods such as random forests will reveal whether the presently unaccounted for deviations from the velocity curve can be predicted in new data. Prior to examining the main research objective, we first validate the ML approach by implementing a regression tree as an alternative to the NLS.[15]

Two approaches were proposed to determine whether an SKU is underperforming, in-line performing, or overperforming relative to the velocity curve. Results indicate the managerially relevant approach with the first class being the top ten percent of SKUs that are located above the velocity curve (overperforming); the second class being the 80% of SKUs that fit the velocity curve (in-line performers); and the third class being the bottom ten percent of SKUs that are located below the velocity curve (underperforming). Using the prediction interval as a threshold results in only 1.7% of the data being outside the prediction interval, and these classes also differ in size and may be problematic for further analysis (see Table 3).

Accuracy, Cohens kappa, and AUC are the selected metrics to evaluate the WRF. Accuracy, Cohens kappa and AUC show acceptable performance (Table 4). Cohen's kappa (0.706) falls into the range of 0.61–0.80, which according to Landis and Koch (1977) shows "substantial" prediction performance. The accuracy and AUC results require a benchmark for comparison. OLR was used as the benchmark and showed weaker performance across all prediction evaluation criteria.

[15] We demonstrate that even a simple ML approach, a regression tree, can achieve comparable results with parametric models. However, in the relationship between distribution and market share, the NLS slightly outperforms the regression tree, suggesting the simple boundary of linear or curvilinear regression models generalizes slightly better. Further details regarding this analysis and results are available in Web Appendix E.

Table 3
Comparison of labeling methods.

SKU class	Prediction interval		Top/bottom ten percent	
	Count	%	Count	%
Overperforming	109	1.2	934	10.0
In-line performing	9,166	98.3	7,453	80.0
Underperforming	46	0.5	934	10.0

Table 4
Prediction metrics for WRF and benchmark OLR on training data.

Criteria	WRF	OLR (benchmark)
Accuracy	0.892	0.805
Cohen's kappa	0.706	0.082
AUC	0.877	0.533

The WRF and OLR (benchmark) models both reasonably predict the three classes of SKUs in the test dataset. The results of the confusion matrix (Table 5) represent across the diagonal the correct predictions, with all other values being incorrect predictions. The WRF correctly identifies ∼83% of the SKUs' performance, compared with ∼37% from the benchmark OLR as per the sensitivity. The specificity indicates a misclassification rate of ∼7% for the WRF, compared with ∼31% from the benchmark OLR.

The WRF provides information about the importance of contributing variables (SKU features). Fig. 4 shows the 30 most important SKU features in descending order, with those of higher importance tending to be closer to the root of the trees. Results show (in descending order of importance) that an SKU's percentage share of medium–large (first ranked) and large stores (third ranked); the average market share deviation of other SKUs for the same brand (second ranked); the average percentage share of category sales of the total revenue of all stores in which the SKU was sold (fourth ranked); and the average number of weeks an SKU was sold in medium–large stores (fifth ranked) are the most important determinants of the WRF prediction performance.

A comparison of variable importance to the benchmark OLR is available in Web Appendix C. The OLR and WRF produce considerably different importance scores for the included variables. The likely reason for the weaker performance on prediction and different importance scores is that the OLR is making strong assumptions about the nature of the relationship between the variables and SKU performance. Whilst OLR can produce reasonable predictions generally, WRF, by making fewer assumptions, has the potential to provide greater insight into the relationship between the variables and their importance to SKU performance. Some observations on the potential reasons for the importance of these variables are now presented.

The first and third-ranked variables are related to the construct of store size, specifically an SKU's share of medium–large and

Table 5
Confusion matrix—correct and incorrect predictions of final WRF and OLR (benchmark) on test data.

Model	SKU class	In-line performing (actual)	Overperforming (actual)	Underperforming (actual)	Average
WRF	In-line performing (predicted)	**1,367**	27	18	
	Overperforming (predicted)	67	**144**	19	
	Underperforming (predicted)	56	15	**149**	
	Sensitivity	0.917	0.774	0.801	0.831
	Specificity	0.879	0.949	0.958	0.929
OLR (Benchmark)	In-line performing (predicted)	**1,483**	174	175	
	Overperforming (predicted)	5	**12**	1	
	Underperforming (predicted)	2	0	**10**	
	Sensitivity	0.995	0.065	0.054	0.371
	Specificity	0.062	0.996	0.999	0.686

The bold values represent the correct predictions.

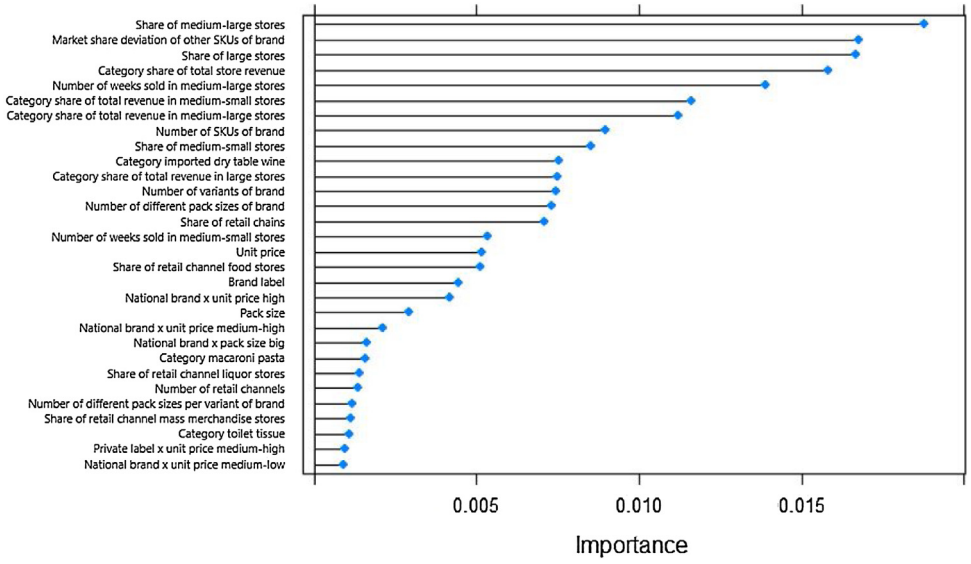


Fig. 4. Variable (SKU feature) importance in the WRF model.

large stores of all the stores in which it is stocked. For a given level of distribution, if an SKU is more likely to be stocked by larger stores, and by implication, less likely to be stocked by smaller stores, the SKU tends to lie below the velocity curve. One possible reason is that SKUs in small stores face fewer competing SKUs (Hristakeva 2020). A retailer's stocking preference may play a role (Alexander 1964; Davies 1994), and may depend on SKU performance as well as store capacities (Shah, Kumar, and Zhao 2015). This phenomenon has been argued to be a fundamental reason for convexity in velocity curves (Hirche et al. 2021, 2019; Farris, Olver, and De Kluyver 1989; Friberg and Sanctuary 2017).

The second-ranked variable indicates the importance of a strong parent brand, expressed as the average market share deviation of other SKUs for the same brand. SKUs tend to be placed above or below the curve depending on the position of the brand's other SKUs relative to the velocity curve. This suggests that common factors such as branding (of NBs and retailer support for own PLs (Farris 2004)), methods of distribution (chain warehouse, wholesalers, or store-door delivery (Waller et al. 2010)), or lagging listing decisions by retailers (Fisher and Vaidyanathan 2014) may explain persistent deviations from category velocity curves.

The fourth-ranked variable, the percentage of category sales in stocking stores, indicates that the SKU is more likely to be available in stores that focus on the particular category, therefore will tend to achieve higher market shares per point of distribution. It also suggests that PCV might be a better measurement choice for velocity curves.

The fifth-ranked variable, the average number of weeks an SKU was sold in medium–large stores, may suggest that supply–demand efficiency and sales consistency play a role in whether an SKU is positioned above or below the velocity curve. This means that avoiding out-of-stock periods and ensuring better shelf-penetration in larger stores may be important factors for underperformance or overperformance. Further, because some SKUs are present in stores for only part of the year, the sales of those "in and out" SKUs will tend to be lower per point of weighted distribution. This might be addressed by shortening the period over which velocity curves are estimated and adjusting the distribution metric to reflect the number of weeks. If category sales are seasonal, further adjustments might be required.

Discussion

In this research, we explore the presently unexplained deviations in the cross-sectional data of the velocity curve. We have extended the research into the convex velocity curve, further validating the findings of Wilbur and Farris (2014) that this relationship extends beyond brands and applies at the individual SKU level. Our study also extends this finding to aggregate distribution across different types of retailers. We determined that on average, the velocity curve accounts for 72% of market share variation for SKUs, highlighting the power of this bi-causal relationship. To account for the unexplained variance, we proposed the adoption of ML as a new methodical approach to examining this relationship.

As a precondition to addressing the research objective we validate that ML achieves similar results to other parametric (nonlinear regression) approaches used to estimate velocity curves for product categories. We find both types of approaches perform equivalently, but are suitable for different needs, in particular when assessing deviations from the velocity curve versus modeling the curve itself. When distribution and market share are continuous variables, the NLS approach only slightly outperforms a simple ML learning approach (regression tree). It appears that for modeling a relationship between two continuous variables, a simple boundary of linear or curvilinear regression model generalizes slightly better than regression trees.

Having established that ML is capable of producing results comparable to the original parametric NLS model, we employed more advanced ML techniques to account for the variation in market share, our pseudo-dependent variable, for which the original bi-causal model of the distribution–market share relationship did not account. We used WRF, a sophisticated ML technique, to predict SKUs that had performance under, in-line with, or over the velocity curve, thus addressing the objective of the research. Previous studies have focused on modeling the velocity curve itself (Kruger and Harper 2006; Reibstein and Farris 1995; Wilbur and Farris 2014), but the incorporation of ML provides substantial insights from class prediction.

The importance of individual variable (feature) inputs in contributing to ML prediction also provides us insight into what may lead to the different performance levels for SKUs. However, we caution that the features driving additional market share are likely to be unstable through time. ML thus represents a useful complementary prediction tool that can be used for scenario analysis with existing assortments/portfolios at a particular point in time, thus supporting continuation/discontinuation decisions or predicting the performance of new SKUs. We found that SKU performance was associated with product line length and type of brand, the overall performance of the parent brand, supply–demand efficiency including sales consistency, as well as store penetration across different store sizes and their category specialization. Considering these findings from our study, we want to reinforce the importance of further developing theory relating to these concepts to fulfil the ideal approach of complementing ML with strong theory (Bradlow et al. 2017). In practice, ML allows automated processing of big and complex data very quickly and provides accurate predictions that can help management to identify opportunities and avoid risks.

Application and Research Directions

The debate about the role of theory for marketing science and strategy is ongoing (Dekimpe 2019; Varadarajan 2019; Bass 1995). Science and technology offer methodical advances that deliver predictions that theory is currently unable to supply. The limitations of theoretical knowledge also limit practical application. The developments in ML and data availability in recent times allow for robust predictive analytics that "makes it possible to identify best courses of action without requiring understanding of the underlying mechanisms" (Bzdok, Altman, and Krzywinski 2018, p. 223). With more data available, classical statistical models become more complex and therefore more difficult to interpret. ML is almost entirely concerned with predictions rather than classical parameter estimations (Mullainathan and Spiess 2017).

However, it is important to highlight that ML is not a substitute for theory, but a complementary tool for better predictions and generating new hypotheses (Cui and Curry 2005). ML, as it stands, is not designed to make statements about causal relationships (nor is classical statistics). Although this is being slowly challenged, see for example, Zhao and Hastie (2019), the innovations are still ad-hoc and do not yet stand on firm theoretical grounds. Bradlow et al. (2017) provide a thorough discussion on how big data and predictive analytics can evolve in retailing. Under ideal circumstances, ML would be used in conjunction with strong theory, as described by Bradlow et al. (2017).

The outcome of this research is relevant for portfolio and category management. There is a need for retail practice to better assess SKU velocity for adoption and delisting decisions (Begoun 2017; Cosgrove 2020; Martin 2014). SKU failure, for example, is a common and expensive problem for CPG suppliers (Fisher and Vaidyanathan 2012). We estimated the general category relationship between distribution and market share for SKUs, and used ML to accurately predict substantive deviations from this estimated relationship. The ML also provides information about the characteristics of the SKUs that may be contributing to their deviation from the velocity curve.

Managers can identify deviating SKUs for further assessment to inform listing and delisting decisions, as well as other SKU decisions (e.g., promotions). For new SKUs, the velocity curve supports the validation of distribution and market share targets by showing whether they are likely to be mutually compatible. For established SKUs, velocity curves show whether SKUs are likely to maintain, increase, or decrease distribution and market share in the future.

Overperforming SKUs are attractive for retailers to add to their category assortments. Therefore, overperformance may require immediate supply considerations for a likely increase in distribution. However, overperforming SKUs may just temporarily reach above average market share (e.g., due to expected or unexpected events, seasonality, or promotion), before dropping back to expected market share levels.

Underperformance of SKUs could be used as a criterion for discontinuing SKUs by retailers and CPG suppliers. Additional promotion efforts may temporarily increase market share of underperforming SKUs, moving them to expected market share levels closer to the curve. Underperformance may be a result of overdistribution, being in too many stores with some not generating sales levels necessary to meet expected market share. On the other hand, underperformance can be a result of recent distribution efforts, potentially resulting in delayed market share uplift.

Our results show that channel- and retail chain composition can be important factors influencing under- and overperformance of SKUs. CPG managers are generally interested in improving the performance of the overall brand and that would include increasing the velocity of the SKUs of that brand on the velocity curve. We would expect that higher velocity results in more retailers adopting the SKU and the movement upward along the curve. On the other hand, SKUs below the curve may be in danger of being discontinued and without understanding why the SKU is below the curve, managers will not be able to determine the urgency of the situation. Once discontinued, improving SKU performance may be much more difficult. It is therefore important that management understands these dynamics, and our research adds to this knowledge and offers a new, additional pathway for new theory.

If needed, further post-hoc analyses such as descriptive analysis, pairwise methods, or controlled experiments might refine the assessment. These will complement but cannot replace the utility of the ML algorithm. This type of post-hoc analysis can give insight into why a particular SKU may be over- or underperforming. The potential reasons for why an SKU over- or underperforms were mentioned in the section regarding the Limitations of Theoretical Development.

Each of these accounts provides testable hypotheses. Initial comparisons to the importance of the variables driving the ML results may indicate which set of hypotheses are strong candidates for subsequent pairwise or experimental tests.

The strength of our ML application is its ability to predict an SKU's performance in new data based on its own, its brand's, and its category's product-related and distribution-related characteristics. This data can be sourced externally and complemented with in-house data. Our results indicate key factors for greater predictive power at a point in time. However, the list of variables (SKU features) that we used to predict SKU performance and explore some of the potential reasons for underperformance or overperformance should be considered at the observational stage of the scientific method for generating formal hypotheses that can be tested in subsequent research.

Researchers can use different metrics and additional inputs, including financial metrics such as return on inventory investment, loyalty metrics, and profitability metrics of customers who buy a particular SKU. It may also be possible to consider velocity effects of overperforming and underperforming SKUs over time. Potential overperformers could benefit from pull effects by retailers (listing decisions), which can subsequently increase distribution and market share. The long-term stability of velocity curves is another potential area for future research. The increased number of SKUs being stocked in multi-channel distribution, may or may not shift the velocity curves over time. Future research may also systematize the possible differences between categories and brand types regarding differences in the velocity curve and deviations from it. There are numerous opportunities to expand this research area.

Appendix A. Summary of variables (feature) input/output

Variable	Type	Number of variables	Formula or condition	Description
Market share (%)	Continuous	1	$\%MS_{iky} = \frac{1}{52}\left(\sum_{w=1}^{52}\%MS_{ikw}\right)$	Average weekly % market share (value) over a 52-week period (year)
Weighted distribution (%Wtd. Dist.)	Continuous	1	$\%Wtd.Dist._{iky} = \frac{1}{52}\left(\sum_{w=1}^{52}\%Wtd.Dist._{iw}\right)$	Average weekly %Wtd. Dist. weighted by annual total revenue of the stores that sell the SKU's category over a 52-week period (year)
Market share deviation (%)	Continuous	1	$MSD_{ik} = MS_{ik} - \frac{\hat{\beta}_{0_k} Wtd.Dist._{ik}^{\hat{\beta}_{1_k}}}{(100-Wtd.Dist._{ik})^{\hat{\beta}_{2_k}}}$	Deviation of market share along the y-axis by subtracting the theoretical market share from its actual observation
SKU performance class	Categorical	1	If i belongs to ten percent with highest MSD_{ik}, then assign 1, if i belongs to ten percent with lowest MSD_{ik}, then assign −1, otherwise 0	3 levels: 0 = normal, 1 = overperforming, −1 = underperforming
Categories	Dummy (0/1)	6 (Tuna category = omitted)	If category k equals k's product_module_code, then assign 1, otherwise 0	0/1 dummy variables for each product category; tuna category is omitted from analysis

Brand label	Categorical	1	If brand description equals "Control Brand" and retailer code is unique, then assign PL, otherwise if retailer code is not unique, then group by retailer code and assign NB	Private label (PL) brands are exclusive to one retail chain; remaining brands are coded as national brands (NBs)
Number of SKUs of brand	Continuous	1	$NSKUB_{b_i ky} = \sum_{SKU=1}^{n} SKU_{b_i ky}$	Number of all individual SKUs of the parent brand to which the SKU belongs
Market share deviation of other SKUs of brand (%)	Continuous	1	$MSD_{kyb_i - i} = \frac{\sum_1^n MS_{kyb_i} - MS_{kyi}}{n_{b_i} - 1_i}$	Average market share deviation from the velocity curve of all other SKUs of the parent brand to which the SKU belongs
Number of variants of brand	Continuous	1	$NVARB_{b_i ky} = \sum_{VAR=1}^{n} VAR_{b_i ky}$	Number of all different variants of the parent brand to which the SKU belongs
Multi-pack	Continuous	1	$multi_{iky} = \sum_{m=1}^{n} Single_{iky}$	Value represents the number of single packs in an individual SKU
Pack size	Categorical	1	$PS_{iky_{1-4}} = SPV_{iky_{1-4}} \times multi_{iky_{1-4}}$	0/1 dummy variable for each pack size level small, medium–small, medium–big, big (1–4); pack size is categorized as small if the pack size (volume) falls in the first quartile of all SKUs in the category; value equals the single package volume multiplied by the number of single items the SKU's package has in the category
Number of different pack sizes per variant of brand	Continuous	1	$NPSVARB_{b_i ky} = \sum_{NPS=1}^{n} NPS_{VARb_i ky}$	Number of different pack sizes per variants of the parent brand to which the SKU belongs
Number of different pack sizes of brand	Continuous	1	$NPSB_{b_i ky} = \sum_{NPS=1}^{n} NPS_{b_i ky}$	Number of different pack sizes of the parent brand to which the SKU belongs
Unit price	Categorical	1	$AUP_{iky_{1-4}} = \frac{APP_{iky_{1-4}}/multi_{iky_{1-4}}}{SPV_{iky_{1-4}}}$	Variable for each unit price level: low, medium–low, medium–high, high (1–4); unit price is categorized as low if the unit price falls in the first quartile of all SKUs in the category; value equals the single package price divided by the single package volume of the SKU in the category
PL × pack size	Dummy (0/1)	3 (small = omitted)	$PLPS = PL_i \times PS_{iky_{1-4}}$	0/1 dummy variable for the interaction of PL and pack size; simple multiplication of the original dummy variables
NB × pack size	Dummy (0/1)	3 (small = omitted)	$NBPS = NB_i \times PS_{iky_{1-4}}$	0/1 dummy variable for the interaction of NB and pack size; simple multiplication of the original dummy variables
PL × unit price	Dummy (0/1)	3 (low = omitted)	$PLAUP = PL_i \times AUP_{iky_{1-4}}$	0/1 dummy variable for the interaction of PL and unit price; simple multiplication of the original dummy variables
NB × unit price	Dummy (0/1)	3 (low = omitted)	$NBAUP = NB_i \times AUP_{iky_{1-4}}$	0/1 dummy variable for the interaction of NB and unit price; simple multiplication of the original dummy variables
Pack size × unit price	Dummy (0/1)	9 (small and low = omitted)	$PSAUP = PS_{iky_{1-4}} \times AUP_{iky_{1-4}}$	0/1 dummy variable for the interaction of pack size and unit prices simple multiplication of the original dummy variables

Number of weeks sold	Continuous	1	$NWS_{iky} = \frac{1}{52}\left(\sum_{w=1}^{52} 1_{ikw}\right)$	Average number of weeks the SKU was sold in any stores that carried the SKU
Number of weeks sold in small stores	Continuous	1	$NWS_{iky_1} = \frac{1}{52}\left(\sum_{w=1}^{52} 1_{ikw_1}\right)$	Average number of weeks the SKU was sold in small stores; stores are categorized as small if their average weekly total revenue falls in the first quartile of all stores selling the category
Number of weeks sold in medium–small stores	Continuous	1	$NWS_{iky_2} = \frac{1}{52}\left(\sum_{w=1}^{52} 1_{ikw_2}\right)$	Average number of weeks the SKU was sold in medium–small stores; stores are categorized as medium–small if their average weekly total revenue falls in the second quartile of all stores selling the category
Number of weeks sold in medium–large stores	Continuous	1	$NWS_{iky_3} = \frac{1}{52}\left(\sum_{w=1}^{52} 1_{ikw_3}\right)$	Average number of weeks the SKU was sold in medium–large stores; stores are categorized as medium–large if their average weekly total revenue falls in the third quartile of all stores selling the category
Number of weeks sold in large stores	Continuous	1	$NWS_{iky_4} = \frac{1}{52}\left(\sum_{w=1}^{52} 1_{ikw_4}\right)$	Average number of weeks the SKU was sold in large stores; stores are categorized as large if their average weekly total revenue falls in the fourth quartile of all stores selling the category
Share of stores by store size (%)	Continuous	4	$\%SSS_{iky_{1-4}} = 100 \times \frac{\sum_{s_{1-4}=1}^{p} Stores_{iksy}}{\sum_{s=1}^{p} Stores_{iksy}}$	SKU's average percentage share of stores by store size (1–4) in which the SKU has been sold; stores are categorized by their average weekly total revenue according to their equivalent quartile of all stores selling the category
Share of retail chains (%)	Continuous	1	$\%SRC_{iky} = 100 \times \frac{\sum_{c=1}^{t} Chains_{ikcy}}{\sum_{c=1}^{t} Chains_{kcy}}$	SKU's average percentage share of retail chains in which SKU has been sold compared with the number of all retail chains that sold the category
Number of retail channels	Continuous	1	$NC_{iky} = \sum_{ch=1}^{5} Channel_{iky}$	Number of different retail channels the SKU has been sold through; Min = 1, Max = 5
Share of retail channels (%)	Continuous	5	$\%C_{iky_{1-5}} = 100 \times \frac{\sum_{s=0}^{p} Channel_{iky_{1-5}}}{\sum_{s=1}^{p} Stores_{iky}}$	A channel's share of all stores the SKU has been sold in; channels include the following: 1 = convenience, 2 = drug, 3 = food, 4 = mass merchandise, 5 = liquor
Category share of total store revenue	Continuous	1	$\%CS_{iky} = 100 \times \frac{\sum_{s=1}^{p} Sales_{ksy}}{\sum_{s=1}^{p} Revenue_{sy}}$	Average percentage share of category sales of the total revenue of all stores in which the SKU has been sold

Category share of total revenue in small stores	Continuous	1	$\%CS_{iky_1} = 100 \times \frac{\sum_{s_1=1}^{p} Sales_{ksy}}{\sum_{s_1=1}^{p} Revenue_{sy}}$	Average percentage share of category sales of the total revenue of all small stores in which the SKU has been sold; stores are categorized as small if their average weekly total revenue falls in the first quartile of all stores selling the category
Category share of total revenue in medium–small stores	Continuous	1	$\%CS_{iky_2} = 100 \times \frac{\sum_{s_2=1}^{p} Sales_{ksy}}{\sum_{s_2=1}^{p} Revenue_{sy}}$	Average percentage share of category sales of the total revenue of all medium–small stores in which the SKU has been sold; stores are categorized as medium–small if their average weekly total revenue falls in the second quartile of all stores selling the category
Category share of total revenue in medium–large stores	Continuous	1	$\%CS_{iky_3} = 100 \times \frac{\sum_{s_3=1}^{p} Sales_{ksy}}{\sum_{s_3=1}^{p} Revenue_{sy}}$	Average percentage share of category sales of the total revenue of all medium–large stores in which the SKU has been sold; stores are categorized as medium–large if their average weekly total revenue falls in the third quartile of all stores selling the category
Category share of total revenue in large stores	Continuous	1	$\%CS_{iky_4} = 100 \times \frac{\sum_{s_4=1}^{p} Sales_{ksy}}{\sum_{s_4=1}^{p} Revenue_{sy}}$	Average percentage share of category sales of the total revenue of all large stores in which the SKU has been sold; stores are categorized as large if their average weekly total revenue falls in the fourth quartile of all stores selling the category

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